

Clustering

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1 Introduction

We discuss about clustering, which is an unsupervised learning technique which identifies groups or clusters of data points.

2 K-means Clustering

2.1 Introduction

K-means Clustering is one of the most simple clustering algorithm. It does not use any probabilistic notions, and only rely on deterministic informations. K-means clustering can be seen as special case of *Gaussian mixture model*, a probabilistic clustering algorithm, which will be shown later.

2.2 Algorithm

Algorithm 1: K-means clustering algorithm

```
Input :  $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$   
         number of clusters  $K$   
         number of iterations  $T$   
  
Output: cluster vectors  $\boldsymbol{\mu}_k$   
         indicator  $\{r_{nk}\}$   
  
// initialize  
1  $\{r_{nk} \leftarrow 0\}$  for  $n \in \{1, \dots, N\}$  and  $k \in \{1, \dots, K\}$   
2 repeat  
   // E step  
3    $\{r_{nk}^{(\text{old})}\}$   
4   for  $n \in \{1, \dots, N\}$  do  
5      $k \leftarrow \arg \min_j \|\mathbf{x}_n - \boldsymbol{\mu}_j\|^2$   
6      $r_{nk} \leftarrow 1$   
7      $\{r_{nk} \leftarrow 0\}$  for  $j \in \{1, \dots, K\}, j \neq k$   
   // M step  
8   for  $k \in \{1, \dots, K\}$  do  
9      $\boldsymbol{\mu}_k \leftarrow \frac{\sum_n r_{nk} \mathbf{x}_n}{\sum_n r_{nk}}$   
10 until  $\{r_{nk}\} = \{r_{nk}^{(\text{old})}\}$  or iterated  $T$  times  
11 return  $\{\boldsymbol{\mu}_k\}, \{r_{nk}\}$ 
```

The algorithm tries to minimize the error function $J = \sum_{n=1}^N \sum_{k=1}^K r_{nk} \|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2$, where r_{nk} is 1 if data point $\mathbf{x}_n$ is included in cluster k , else 0. and $\boldsymbol{\mu}_k$ are the 'prototype' of the cluster vector, representing the center of the clusters. The algorithm first minimize the loss function about r_{nk} with $\boldsymbol{\mu}_k$ fixed, which is simply

$$r_{nk} = \begin{cases} 1 & \text{if } k = \arg \min_j \|\mathbf{x}_n - \boldsymbol{\mu}_j\|^2 \\ 0 & \text{otherwise} \end{cases}$$

This is **E step** if we look it as a kind of EM algorithm. Then, the algorithm fixes r_{nk} and minimize about $\boldsymbol{\mu}_k$. The objective function J is a quadratic function of $\boldsymbol{\mu}_k$ and can be minimized easily, with

$$\boldsymbol{\mu}_k = \frac{\sum_n r_{nk} \mathbf{x}_n}{\sum_n r_{nk}}.$$

We can derive a sequential version of this batchwise algorithm.

Algorithm 2: Sequential K-means clustering algorithm

Input : $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$
 number of clusters K
 number of iterations T
Output: cluster vectors $\boldsymbol{\mu}_k$
 indicator $\{r_n\}$

```
// initialize
1  $\{r_n \leftarrow 0\}$  for  $n \in \{1, \dots, N\}$  and  $k \in \{1, \dots, K\}$ 
2  $\{N_k \leftarrow 0\}$ 
3 foreach  $\mathbf{x} \in \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$  do
4    $k \leftarrow \arg \min_j \|\mathbf{x}_n - \boldsymbol{\mu}_j\|^2$ 
5    $N_k \leftarrow N_k + 1$ 
6    $r_n \leftarrow k$ 
7    $\boldsymbol{\mu}_k \leftarrow \boldsymbol{\mu}_k + \frac{1}{N_k}(\mathbf{x}_n - \boldsymbol{\mu}_k)$ 
8 return  $\{\boldsymbol{\mu}_k\}, \{r_n\}$ 
```

The sequential k-means algorithm does not change the data points which are already clustered, unlike the original k-means algorithm, and the indicator changed from one-hot encoding to integer value.

2.3 Applications

K-means clustering can be used at image segmentation, specifically color segmentation.

3 Mixtures of Gaussians

The **Gaussian mixture model** (GMM) is a simple linear superposition of Gaussian components. This can be formulated in terms of discrete latent variables. See this [document](#) for more information.